

Chapter 2 C Fundamentals

Programming Projects[1]

1. Write a program that uses `printf` to display the following picture on the screen:

```
      *
     *
    *
   * *
  * *
 *

```

Answer: [Source Code](#)

2. Write a program that computes the volume of a sphere with a 10-meter radius, using the formula $v = \frac{4}{3}\pi r^3$. Write the fraction $\frac{4}{3}$ as `4.0f/3.0f`. (Try writing it as `4/3`. What happens?) *Hint:* C doesn't have an exponentiation operator, so you'll need to multiply r by itself twice to compute r^3 .

Answer: [Source Code](#)

3. Modify the program of Programming Project 2 so that it prompts the user to enter the radius of the sphere.

Answer: [Source Code](#)

4. Write a program that asks the user to enter a dollars-and-cents amount, then displays the amount with 5% tax added:

```
Enter an amount: 100.00
With tax added: $105.00

```

Answer: [Source Code](#)

5. Write a program that asks the user to enter a value for x and then displays the value of the following polynomial:

$$3x^5 + 2x^4 - 5x^3 - x^2 + 7x - 6$$

Hint: C doesn't have an exponentiation operator, so you'll need to multiply x by itself repeatedly in order to compute the powers of x . (For example, `x*x*x` is x cubed.)

Answer: [Source Code](#)

Bibliography

- [1] K. N. King, *C Programming A Modern Approach*, 2nd ed. 2008.